

C# Basics Cheat Sheet (Memory-Friendly)

1. What is a Programming Language?

- A programming language lets humans write instructions for a computer.
- The instructions are called **source code** (or simply **code**).
- Different languages have different **syntax**, but many programming concepts are similar.

Example:

```
Console.WriteLine("Hello World!");
```

2. What is Compilation?

- The computer **cannot understand source code directly**.
- A **compiler** converts source code into **machine code** (binary instructions) that the CPU can execute.

Flow:

```
Source Code
  ↓
Compiler
  ↓
Machine Code
  ↓
CPU Executes
```

Remember:

Write Code → Compile → Run

3. What is Syntax?

Syntax = The rules for writing C# code.

Just like English has grammar rules, C# has syntax rules.

Examples:

- Correct capitalization
- Parentheses ()
- Quotes " "
- Semicolon ;

Wrong syntax causes **compile errors**.

4. Syntax Highlighting

The code editor changes colors to:

- Keywords
- Strings
- Classes
- Methods

This helps find mistakes quickly.

5. Understanding

```
Console.WriteLine("Hello World!");
```

Break it down:

Console



Class

.



Member Access Operator

WriteLine()



Method

("Hello World!")



Input Parameter

;



End of Statement

6. Literal String

Anything inside double quotes is called a **Literal String**.

Example:

"Hello"

"Safin"

"Programming"

7. Class

A **Class** contains methods.

Example:

Console

Think of a class as a **toolbox**.

8. Member Access Operator

The **dot (.)** is called the **Member Access Operator**.

It connects a class to one of its methods.

Example:

Console.WriteLine();

Console

.

WriteLine

9. Method

A **Method** performs an action.

You can recognize a method because it always has:

()

Example:

WriteLine()

ReadLine()

Clear()

10. Method Invocation Operator

The parentheses:

()

are called the **Method Invocation Operator**.

Even if there are no parameters, you **must** include the parentheses.

Correct:

```
Console.Clear();
```

Wrong:

```
Console. Clear;
```

11. Input Parameter

The value inside the parentheses is called an **Input Parameter**.

Example:

```
Console.WriteLine("Hello");
```

Input Parameter:

```
"Hello"
```

12. Semicolon (;)

The semicolon means:

End of Statement

Every complete C# statement usually ends with ;

Correct:

```
Console.WriteLine("Hello");
```

Wrong:

```
Console.WriteLine("Hello")
```

13. Statement

A **Statement** is one complete instruction.

Example:

```
Console.WriteLine("Hello");
```

14. Important Capitalization

Correct:

```
Console.WriteLine()
```

Wrong:

```
console.writeline()
```

```
Console.writeline()
```

```
console.WriteLine()
```

C# is **case-sensitive**.

15. Things to Always Remember

✓ Use

```
Console.WriteLine("Message");
```

✓ Use double quotes for text

```
"Hello"
```

✓ End with

```
;
```

✓ Methods are always used

```
()
```

✓ Capitalize correctly

```
Console  
WriteLine
```

16. Memory Formula

Programming Language



Write Source Code



Compiler



Machine Code



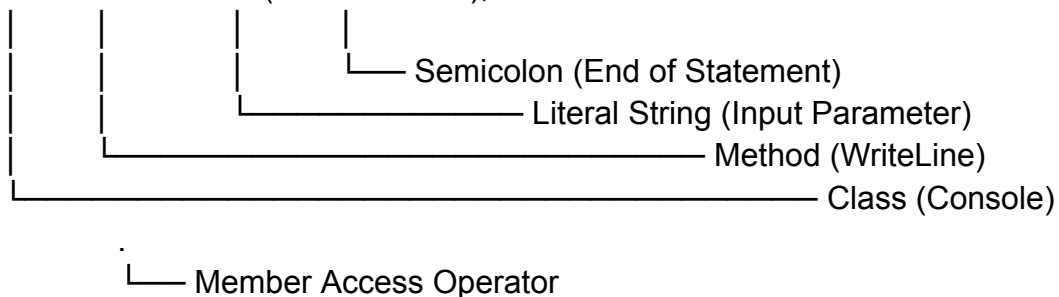
CPU Runs Program

17. Quick Revision Table

Term	Meaning	Example
Programming Language	Used to write instructions	C#
Source Code	Code written by a developer	<code>Console.WriteLine();</code>
Compiler	Converts code into machine code	C# Compiler
Compilation	Process of converting code	Compile → Run
Syntax	Rules of writing code	Correct punctuation
Class	Contains methods	<code>Console</code>
Member Access Operator	Connects class to method	<code>.</code>
Method	Performs an action	<code>WriteLine()</code>
Method Invocation Operator	Parentheses used to call methods	<code>()</code>
Input Parameter	Data passed to a method	<code>"Hello"</code>
Literal String	Text inside quotes	<code>"World"</code>
Statement	One complete instruction	<code>Console.WriteLine("Hi");</code>
Semicolon	Ends a statement	<code>;</code>

18. One-Line Summary

`Console.WriteLine("Hello World!");`



Easy mnemonic: C.C.M.P.S.

- **C** = Class (**C**onsole)
- **C** = Connect with. (member access)
- **M** = Method (**W**riteLine())
- **P** = Parameter ("**H**ello **W**orld!")
- **S** = Semicolon (;)